

Noosphere Technical Reader

LLMs, Embedding Geometry, NLP Extraction, Knowledge Graphs, and Coherence-Based
Intellectual Capital Modeling

Prepared for Michael Quintin / Theseus

May 17, 2026

Contents

1	What This Packet Is	4
1.1	The Four Substrates	4
2	Noosphere as a System	4
2.1	The Actual Data Shape	4
2.2	Operational Pipeline	5
3	LLMs: From Next Tokens to Reasoning Instruments	5
3.1	The Basic Transformer Picture	5
3.2	Why Scale Matters	5
3.3	Alignment Changes the Instrument	6
4	Embeddings: Semantic Objects in Vector Space	6
4.1	From Word2vec to Sentence Embeddings	6
4.2	Cosine Is Useful but Not Enough	6
4.3	Manifolds, Symmetry, and Linear Directions	6
5	The Noosphere Geometry Hypothesis	7
5.1	What the Internal Evidence Actually Says	7
6	NLP Extraction: Turning Text into Claims	7
6.1	Classical Pipeline	7
6.2	Modern Extraction	8
6.3	Argument Mining	8
7	Knowledge Graphs and Cascade Structure	8
7.1	Why a Graph Is Necessary	8
7.2	KG Embeddings	8
8	Logic, Contradiction, and Belief Revision	9
8.1	Classical Logic Is Not Enough	9
8.2	Bayesian Derived Views	9
9	Coherence: Philosophy and Computation	9
9.1	Coherence as Justification vs Coherence as Truth	9
9.2	Six-Layer Coherence	9

10 From Reasoning to Decisions	10
10.1 Trace Before Prose	10
10.2 The Correct Liability Ladder	10
11 Frontier Research Map	10
11.1 Mechanistic Interpretability	10
11.2 Representation Engineering	11
11.3 KG and LLM Convergence	11
12 A One-Go Reading Path	11
12.1 Pass 1: Build the Skeleton	11
12.2 Pass 2: Understand Noosphere’s Moving Parts	11
12.3 Pass 3: Enter the Frontier	11
12.4 Pass 4: Build the Critique	12
13 Concept Glossary	12
14 Deep Dive I: Transformer Mechanics	13
14.1 Tokenization Is Already a Modeling Choice	13
14.2 Embeddings and Positional Information	13
14.3 Self-Attention in One Equation	13
14.4 Multi-Head Attention	13
14.5 Training Objective vs. User Intent	14
14.6 Inference-Time Behavior	14
15 Deep Dive II: Embedding Geometry Mechanics	14
15.1 Distributional Semantics	14
15.2 SVD, Factorization, and Latent Axes	14
15.3 The Difference Between Similarity and Relation	14
15.4 High-Dimensional Geometry Has Traps	15
15.5 Hoyer Sparsity	15
15.6 Householder Reflection	15
16 Deep Dive III: Truth, Linear Representations, and Steering	15
16.1 Linear Representation Hypothesis	15
16.2 Truth Directions	16
16.3 Representation Engineering	16
16.4 Superposition as a Warning	16
17 Deep Dive IV: Extraction Engineering	16
17.1 The Extraction Contract	16
17.2 Chunking and Span Selection	17
17.3 Coreference and Entity Hygiene	17
17.4 Relation Extraction	17
17.5 Prompting and RAG	18
18 Deep Dive V: Graphs, Ontologies, and Cascade Reasoning	18
18.1 Property Graph vs. Formal Ontology	18
18.2 Edge Semantics	18
18.3 Cycle Handling	19
18.4 Knowledge Graph Embeddings	19

19 Deep Dive VI: Coherence, Probability, and Critique	19
19.1 Coherence as Constraint Satisfaction	19
19.2 Coherence Is Not Probability	19
19.3 Paraconsistency and Non-Explosion	19
19.4 Belief Revision	20
20 Deep Dive VII: Verification and Method Discipline	20
20.1 What Must Be Versioned	20
20.2 What Counts as Evidence for the Quintin Hypothesis	20
20.3 Decision-Trace Verification	21
21 Exercises for Rebuilding Understanding	21
21.1 Exercise 1: Re-derive Attention	21
21.2 Exercise 2: Show the Cosine Paradox	21
21.3 Exercise 3: Build a Tiny Cascade	22
21.4 Exercise 4: Coherence vs. Truth	22
21.5 Exercise 5: Write a Decision Trace	22
22 Source Inventory	22
23 Closing Synthesis	26

1 What This Packet Is

This reader turns the linked technical reading list into a single continuous study document. It is not a pasted anthology. Most of the source material is copyrighted even when it is publicly readable, so the main body is a synthesized walk-through: it teaches the ideas, cites the source trail, and tells you where the official PDF or page was cached when available. The source cache and the manifest live beside this PDF.

Core claim

Noosphere is easiest to understand as a pipeline that turns text into typed propositions, embeds those propositions into vector space, places them in a graph of support and conflict, and then asks whether the resulting structure is coherent enough to support a conclusion or decision.

1.1 The Four Substrates

The source guide divides the background into four domains, and that division is correct:

1. **The LLM substrate.** Claude or another model extracts claims, distills principles, judges coherence layers, and writes explanations. The model is not a neutral parser. Its transformer architecture, alignment training, and prompt interface shape the outputs.
2. **The embedding substrate.** Claims and principles become high dimensional vectors. Noosphere's geometric hypothesis is that relations such as similarity, contradiction, truth, and coherence leave measurable structure in those vectors or in differences between them.
3. **The NLP extraction substrate.** Raw text must be segmented, attributed, disambiguated, and transformed into atomic truth-apt claims before a graph can be trusted.
4. **The graph and logic substrate.** The knowledge store is not only a bag of embeddings. It is a graph of claims, conclusions, supports, refutations, dependencies, revisions, and derived probabilistic views.

2 Noosphere as a System

The local Theseus repository defines the project as a machine for extracting principles from a curated corpus, running algorithms over those principles, and making conclusions or bets accountable to downstream evidence. In repository terms, **theseus-codex** is the workspace and public face, **noosphere** is the reasoning engine, and **dialectic** is the live conversation analyzer.

2.1 The Actual Data Shape

The fundamental unit in **noosphere/noosphere/models.py** is a **Claim**: an atomic proposition extracted from text, attributed to a speaker, and optionally positioned in embedding space. Relations between claims and principles include support, contradiction, refinement, instantiation, extension, analogy, presupposition, and qualification.

Why this matters

If claim extraction is wrong, every downstream graph and coherence score is contaminated. If embeddings are stale, geometry-based scores drift. If graph edges are under-specified, the public explanation can look rigorous while the actual decision trace is underspecified. Noosphere's engineering problem is to make each of those failure modes visible.

2.2 Operational Pipeline

The software path can be read as:

1. **Source ingestion.** Text enters as chunks with metadata and provenance.
2. **Claim extraction.** `ClaimExtractor` asks an LLM for JSON containing truth-apt claims, claim type, hedges, evidence pointers, and whether the author actually endorses the claim.
3. **Principle extraction.** Later extraction turns source spans into principle-shaped decision rules with domains of applicability, proxies, and examples.
4. **Embedding.** `embedding_pipeline.py` idempotently embeds source text, records the model name and text hash, writes vectors into the store, and updates locality indices for claims and conclusions.
5. **Graph construction.** The cascade graph records nodes and edges such as supports, refutes, contradicts, depends on, specializes, generalizes, coheres with, predicts, and extracted from.
6. **Coherence and contradiction.** Pairwise checks and contradiction engines score whether claims cohere, conflict, or remain unresolved.
7. **Trace generation.** Algorithmized decisions must record named metrics, rule graph nodes, vetoes, and outputs. The prose explanation is secondary; the trace is the source of truth.

3 LLMs: From Next Tokens to Reasoning Instruments

3.1 The Basic Transformer Picture

The transformer is a differentiable architecture for contextual sequence processing. Tokens enter as vectors. Each layer lets tokens read from other tokens through attention, then transforms the result through feed-forward networks. The original architecture is in [Vaswani et al.](#); Jay Alammar’s [Illustrated Transformer](#) is still the best first visual pass; Karpathy’s GPT walkthrough is the best bridge from diagram to executable code.

The minimum mental model is:

- A token embedding is a learned vector representing a symbol in context.
- Self-attention computes weighted mixtures of other token representations.
- Query, key, and value projections are learned views of each token. Query and key determine which tokens attend to each other; value carries the content to mix.
- Layer normalization, residual connections, and feed-forward blocks make deep composition trainable.
- A language model is trained to predict the next token, but the internal representations needed to do this become useful general-purpose features.

3.2 Why Scale Matters

Kaplan et al. made scaling laws a central part of LLM engineering: loss changes predictably with model size, dataset size, and compute. Hoffmann et al.’s Chinchilla work corrected the naive lesson by showing that more parameters are not enough; compute-optimal training requires enough data for the model size. For Noosphere, the point is not merely that larger models are better. It is that model behavior is partly a function of scaling regime, not only prompt wording.

This matters for intellectual-capital modeling because a claim extractor using a frontier model and one using a small local model are not just two implementations of the same parser. They may differ in entity resolution, implicit premise recovery, sensitivity to hedges, and willingness to infer structure not explicitly stated.

3.3 Alignment Changes the Instrument

Instruction tuning and RLHF, as in the InstructGPT paper, alter the interface between raw next-token prediction and user-facing behavior. Constitutional AI adds another layer: model behavior is shaped by critique, revision, and principle-guided feedback. That is useful, but it means a model used inside Noosphere carries inherited priors about harmlessness, helpfulness, refusal, uncertainty, and argumentative framing.

Practical implication

When Claude extracts a claim or judges a contradiction, it is not an inert semantic microscope. It is an aligned language model. Noosphere should therefore log prompts, model versions, schema versions, and failure modes so that later methodology review can distinguish source truth from model-induced structure.

4 Embeddings: Semantic Objects in Vector Space

4.1 From Word2vec to Sentence Embeddings

Word2vec made the modern intuition vivid: learned vectors can encode semantic regularities. GloVe made the co-occurrence matrix factorization view explicit. Jurafsky and Martin’s Chapter 6 is the clean textbook path through PPMI, SVD, word2vec, GloVe, and evaluation. Sentence-BERT then moves the operational unit from isolated words to whole sentences or short paragraphs, which is closer to Noosphere’s claims and principles.

The crucial shift is this: once propositions are vectors, reasoning systems can query neighborhoods, compare directions, estimate distances, and cluster meaning. But every one of those operations depends on the embedding model, the text span, the normalization, and the metric.

4.2 Cosine Is Useful but Not Enough

Cosine similarity measures angle, not logical relation. Two contradictory sentences can have high cosine similarity because they discuss the same topic with almost the same words. This is the “cosine paradox” behind Theseus’s internal contradiction-geometry work. If contradiction is topic-local, then raw similarity can confuse opposition with agreement.

Noosphere’s geometry rationale therefore looks at **difference vectors** and **Hoyer sparsity**. If two claims differ along a small number of meaning-bearing dimensions, the difference vector may concentrate mass in those dimensions. Hoyer sparsity is a scalar in $[0, 1]$ that is low for dense vectors and high for vectors whose mass is concentrated.

4.3 Manifolds, Symmetry, and Linear Directions

Bronstein et al.’s geometric deep learning monograph broadens the picture: deep learning architectures can be read as ways to exploit symmetries in the domain. Convolution exploits translation symmetry on grids. Graph neural networks exploit permutation structure on graphs. Transformers can be read in the same general family of symmetry-aware architectures.

The manifold hypothesis, treated rigorously by Fefferman, Mitter, and Narayanan, supplies a related intuition: high-dimensional data often lies near a lower-dimensional structure. For Noosphere, the speculative version is that “meaning” may occupy structured regions inside an embedding or activation space rather than filling the ambient dimension uniformly.

The Linear Representation Hypothesis and work such as Geometry of Truth, Contrast-Consistent Search, Inference-Time Intervention, and Representation Engineering then ask whether concepts and truth-related properties can be represented as directions. If this is right, a coherence or contradiction

detector can search for directions or subspaces. If it is wrong or only locally true, direction-based methods will overstate what the geometry can support.

5 The Noosphere Geometry Hypothesis

The local contradiction engine records the strongest version of the firm’s geometric bet. It estimates a learned contradiction direction, reflects one embedding across a hyperplane using a Householder transform,

$$b' = b - 2(b \cdot \hat{d})\hat{d},$$

then scores Hoyer sparsity of $b' - a$. The method version is recorded as `geometry/householder/v2`, with a reliability curve and confidence band.

This is an ambitious hypothesis. It is not yet a proven theorem or a mature benchmark result.

5.1 What the Internal Evidence Actually Says

The local QH benchmark results are deliberately important because they prevent the reader from absorbing only the optimistic part of the source list. The first QH-v1 run reported that the firm contradiction-geometry probe lost on overall 3-way accuracy to cosine and even to a random baseline, while still showing a better AUROC slice for contradicting versus coherent. The Householder ablation then found a zero-power label test under the frozen threshold: all variants predicted the same label, so the run could neither justify removing nor confirming the Householder step.

Rigorous reading

The current evidence supports continued experimentation, not triumphal claims. The geometry may contain useful signal, but the production argument must pass through calibration, benchmark design, ablations with power, and cross-model replication before it can be treated as a reliable contradiction detector.

6 NLP Extraction: Turning Text into Claims

6.1 Classical Pipeline

Noosphere needs more than embeddings. It must decide what the text says, who says it, whether the author endorses it, and how claims relate. Classical NLP still matters here:

- Tokenization and sentence segmentation decide the units of analysis.
- Part-of-speech and dependency parsing expose grammatical structure.
- Named-entity recognition finds people, organizations, concepts, and places.
- Coreference resolution decides when “the founder,” “he,” and a proper name refer to the same entity.
- Relation extraction converts text into typed edges.

spaCy and NLTK are included in the source guide because even an LLM-backed system benefits from knowing what older pipeline components do. Jurafsky and Martin’s information-extraction chapter is the cleanest compact technical reference.

6.2 Modern Extraction

BERT shifted extraction by making pretrained contextual encoders standard. REBEL and neural relation extraction show how relation triples can be generated or classified from text. LLM-based extraction adds flexibility but also danger: the model can infer plausible structure that was not asserted, merge distinct claims, lose hedges, or fail to preserve speaker attribution.

The local **ClaimExtractor** is designed around exactly that risk. Its prompt requires JSON, distinguishes author assertions from external or opposing claims, and attaches hedges and evidence pointers. It explicitly tries to avoid polluting founder beliefs with prompts, counterpositions, or rhetorical questions that the author is merely engaging.

6.3 Argument Mining

Argument mining is the closest established field to Noosphere’s higher-level principle extraction. Stab and Gurevych’s work on claims, premises, and support relations matters because Noosphere’s graph needs typed argumentative edges, not just semantic similarity. A good extraction system must know the difference between:

- a claim,
- evidence for a claim,
- a premise in an argument,
- a qualification,
- an objection,
- an example, and
- a stronger principle abstracted from several claims.

7 Knowledge Graphs and Cascade Structure

7.1 Why a Graph Is Necessary

Embeddings give soft geometry. Graphs give inspectable structure. A knowledge graph can represent identity, provenance, support, contradiction, abstraction, and dependency. Hogan et al. provide the reference survey; NetworkX supplies the implementation vocabulary; Diestel supplies the formal graph-theory background.

The local **CascadeGraph** is not a generic graph toy. It enforces conflict rules, prevents cycles for dependency edges, attaches method invocation provenance to edges, and supports deterministic projection into a Bayesian skeleton.

7.2 KG Embeddings

TransE begins with a simple geometric idea: a relation can act like a translation from head entity to tail entity. Later KG-embedding models complicate this with different spaces, scoring functions, rotations, complex embeddings, and auxiliary text. For Noosphere, the point is less that TransE should be copied and more that graph structure and vector structure can be coupled.

This is the natural bridge between a symbolic graph of claims and an embedding space of meanings. It also introduces a warning: graph embeddings optimize for link prediction, not necessarily for philosophical coherence or truth.

8 Logic, Contradiction, and Belief Revision

8.1 Classical Logic Is Not Enough

If a system stores live human beliefs, contradictions will appear. Classical logic contains the principle of explosion: from a contradiction, anything can follow. That is unacceptable for a practical knowledge system. Paraconsistent logic is therefore not an exotic side topic; it is one way to reason in a graph that can contain unresolved conflict without collapsing.

Mereology matters because principles can be composed from parts and wholes. Belief revision matters because source retractions and new evidence should change stored belief in a controlled way. Description logic matters if the ontology becomes formal enough to use OWL-like classes, subsumption, and satisfiability.

8.2 Bayesian Derived Views

The local Bayesian Belief Layer makes the same distinction carefully: the cascade remains the primary representation, while the Bayesian network is a derived view rebuilt on demand. It turns an acyclic skeleton into binary truth-valued nodes with conditional probability tables, then supports marginal probabilities, evidence updates, and sensitivity analysis.

That is exactly the right caution. A posterior probability is useful only when the model structure and CPTs are honest about what they assume. Noosphere’s specification keeps the BN founder-side, reports exact versus approximate inference, and distinguishes seeded rows from data-fit rows.

9 Coherence: Philosophy and Computation

9.1 Coherence as Justification vs Coherence as Truth

The Stanford Encyclopedia entries separate two claims that are easy to conflate. Coherentism about justification says a belief is justified by its role in a mutually supporting system. The coherence theory of truth says truth itself is a matter of coherence among beliefs. Noosphere should be extremely careful here.

A coherence score may be evidence that a belief fits the firm’s corpus. It is not automatically evidence that the belief is true. Bovens and Hartmann’s work, Olsson’s critique of coherentism, and Thagard’s computational coherentism all push against naive equations of coherence with probability or truth.

9.2 Six-Layer Coherence

The local six-layer method combines:

1. NLI consistency,
2. abstract argumentation,
3. probabilistic coherence,
4. embedding geometry,
5. information-theoretic compression, and
6. an LLM judge.

The aggregator requires a 4-of-6 supermajority for a definitive cohere or contradict verdict. That design is sensible as a hedge against any single weak signal, but the local rationale correctly names the weakness: layers are not fully independent. NLI and geometry can share embedding failure modes; the probabilistic layer and LLM judge can share model-induced reasoning failure modes. Supermajority only helps when errors are not too correlated.

10 From Reasoning to Decisions

10.1 Trace Before Prose

The Algorithmized Decision Making document states the right contract: prose is generated from the trace, not the other way around. The trace must contain named metrics, typed values, low-confidence flags, method versions, rule graph nodes, vetoes, and the final output.

For intellectual-capital modeling, this is the difference between an essay that sounds rigorous and a decision that can be audited. A forecast or investment candidate should be reconstructible from:

- the market or question,
- the event representation,
- the corpus slice,
- the active principles,
- calibration state,
- liquidity/cost data where relevant,
- metric values,
- rule graph version, and
- vetoes considered.

10.2 The Correct Liability Ladder

Theseus’s architecture draws a useful ladder:

Layer	Output	Primary judge
Interpretive reasoning	Essay or conclusion	Reader review
Decision trace	Data structure	Algorithmic audit
Investable output	Action or abstention	Market outcome

Each step removes prose freedom and increases accountability. That is the engineering core of “intellectual capital modeling”: principles are not just stored ideas; they become constrained inputs to traceable decisions.

11 Frontier Research Map

11.1 Mechanistic Interpretability

Transformer circuits work asks what computation the model actually performs. Superposition says features may share neurons or directions, which makes clean linear interpretations dangerous. Sparse autoencoder work tries to recover more monosemantic features. Induction-head work is important because it is one of the few cases where a behavior has been tied to identifiable mechanism.

For Noosphere, this literature supplies both hope and constraint. Hope: models do contain internal structure that can sometimes be decomposed. Constraint: surface embeddings and activation directions can be polysemantic, basis- dependent, layer-dependent, and prompt-sensitive.

11.2 Representation Engineering

Representation engineering treats activations as a control surface: identify directions, test whether they correspond to concepts, and intervene. This is adjacent to the Quintin Hypothesis because both ask whether abstract semantic properties have geometric structure. The difference is that representation engineering usually works inside model activations, while many Noosphere methods operate on external text embeddings or stored claim vectors.

11.3 KG and LLM Convergence

The 2023-2025 KG/LLM survey literature frames three broad patterns:

- KGs can enhance LLMs by supplying explicit facts, retrieval context, and interpretability handles.
- LLMs can augment KGs by extracting entities, relations, ontologies, and candidate edges.
- Hybrid systems can use LLMs and KGs in a loop, where explicit graph state and parametric model state correct each other.

Noosphere is squarely in the hybrid category. Its risk is also the hybrid category’s risk: generated graph edges can look explicit while inheriting hidden LLM uncertainty, and LLM answers can look grounded while leaning on parametric knowledge not present in the graph.

12 A One-Go Reading Path

12.1 Pass 1: Build the Skeleton

1. 3Blue1Brown Essence of Linear Algebra.
2. Alammr’s Illustrated Transformer.
3. Karpathy’s GPT-from-scratch video.
4. Jurafsky and Martin Chapter 6.
5. Hogan et al. Knowledge Graphs, introductory sections.

12.2 Pass 2: Understand Noosphere’s Moving Parts

1. Read the local Theseus README.
2. Read `claim_extractor.py` and the source manifest notes.
3. Read `embedding_pipeline.py`.
4. Read the six-layer coherence rationale.
5. Read the contradiction-geometry rationale, then the QH benchmark and Householder ablation summaries.
6. Read Algorithmized Decision Making, especially the trace/rule-graph contract.

12.3 Pass 3: Enter the Frontier

1. Linear Representation Hypothesis.
2. Geometry of Truth.
3. Discovering Latent Knowledge without Supervision.

4. Inference-Time Intervention.
5. Representation Engineering.
6. Toy Models of Superposition.
7. Scaling Monosemanticity.

12.4 Pass 4: Build the Critique

1. SEP on coherentist justification and coherence theory of truth.
2. Paraconsistent logic.
3. Belief revision.
4. Thagard’s coherence-as-constraint-satisfaction work.
5. Bovens, Hartmann, Olsson, and BonJour if you have library access.

13 Concept Glossary

Activation A vector inside a neural network layer, often more directly analyzable than the final output.

Attention A learned operation letting one token representation read from others.

Cascade graph Theseus’s primary support/refutation/dependency structure over claims, conclusions, principles, and artifacts.

Claim An atomic truth-apt proposition extracted from text.

Coherence Fit among claims in a system. It may justify a belief without proving it true.

Contradiction direction A learned or fallback vector estimating where an embedding’s negating counterpart should lie.

Embedding A vector representation of text or another object.

Hoyer sparsity A scalar measuring whether vector mass is concentrated in few coordinates.

Knowledge graph A graph representation of entities, relations, context, identity, and provenance.

LLM alignment Training and feedback processes that make a model follow instructions and norms rather than merely continue text.

Paraconsistent logic A logic family that can tolerate contradictions without deriving everything.

Principle A reusable decision rule or general claim distilled from source material.

Representation engineering Identifying and intervening on model activation directions associated with concepts.

Rule graph A versioned graph of thresholds, combiners, buckets, and vetoes used to turn metrics into decisions.

Superposition The phenomenon where multiple features share representational dimensions.

14 Deep Dive I: Transformer Mechanics

14.1 Tokenization Is Already a Modeling Choice

Before a transformer sees text, the text is broken into tokens. A token may be a whole word, a word piece, punctuation, or a byte-level fragment. This matters because the model does not reason over words as humans encounter them on the page. It reasons over token ids that index rows in an embedding table. A rare technical word may split into several fragments; a common phrase may receive a more stable representation because the model has seen its pieces together many times.

For Noosphere, tokenization matters in three places. First, extraction prompts with unusual jargon can produce brittle outputs if the model has only a weak representation of the terms. Second, evidence pointers can drift when the model summarizes a span whose token boundaries do not match human sentence boundaries. Third, very short claims can be embedding-poor: their vectors may be dominated by broad topical features rather than precise logical content.

14.2 Embeddings and Positional Information

The initial representation of a token is a vector. A transformer also needs some way to represent order. The original transformer used sinusoidal positional encoding; later models use learned or rotary variants. The common point is that the model must distinguish "A caused B" from "B caused A." Noosphere's claim extraction depends heavily on that distinction, especially when it tries to extract causal chains, support relations, or contradiction relations.

14.3 Self-Attention in One Equation

For a matrix of token representations X , attention constructs:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

The attention weights are:

$$\text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right),$$

and the output is those weights multiplied by V . In plain terms: each token forms a query, each token exposes a key, and the dot products decide how much information to read from each value vector.

This is why attention is not just lookup. It is content-dependent routing. In a claim like "The policy reduced inflation, but only because demand collapsed," later tokens can route attention back to "but only because" and change the interpretation of the apparent causal claim. A bag-of-words model would struggle with this; a transformer has a mechanism that can represent it.

14.4 Multi-Head Attention

One attention head can learn one routing pattern. Multi-head attention lets different heads learn different relations at the same layer: syntax, entity tracking, negation, quotation, induction, formatting, or task-specific prompt structure. Interpretability work is interested in heads because some heads appear to implement recognizable algorithms, while others remain diffuse.

Noosphere should care about this because the same prompt can elicit different behaviors depending on which internal circuits are activated. A prompt that asks for "claims" might trigger summarization habits unless the schema and examples strongly distinguish claims from premises, questions, quotations, and objections.

14.5 Training Objective vs. User Intent

The base training objective is next-token prediction. That objective does not directly say "extract only author-endorsed claims" or "do not invent an edge in the graph." The model learns a broad conditional distribution over text. The instruction-tuned interface then adds a behavioral layer that makes the model follow user requests. RLHF and constitutional methods add yet more pressure toward helpfulness, safety, and stylistic compliance.

That stack creates a useful but dangerous instrument. It can parse, explain, and generalize. It can also over-help by filling gaps. In Noosphere, over-help is not benign: an invented relation can become a graph edge, and a graph edge can become evidence in a decision trace.

14.6 Inference-Time Behavior

At inference time, the model samples or selects the next token repeatedly. Even with temperature near zero, deterministic behavior is not the same as truth-preserving behavior. The model may be deterministic about a wrong extraction if the prompt under-specifies the schema. That is why Noosphere's best design pattern is not "ask the model and trust it"; it is "ask the model inside a typed contract, validate the result, record the version, and test the failure mode."

15 Deep Dive II: Embedding Geometry Mechanics

15.1 Distributional Semantics

The oldest lesson in modern embeddings is distributional: words or spans that occur in similar contexts tend to acquire similar representations. Count-based methods made this explicit with co-occurrence matrices. Neural methods learn the representation through prediction. The result is not a dictionary definition. It is a compressed statistical position inside a learned space.

This is why embeddings are powerful for retrieval. If two claims use different surface words but occupy similar contexts, embeddings can bring them close. It is also why embeddings are risky for logic. Logical negation can preserve most context while reversing truth conditions.

15.2 SVD, Factorization, and Latent Axes

Singular value decomposition gives a clean mathematical model for why latent semantic axes can emerge. A large co-occurrence matrix can be approximated by lower-rank factors. Those factors are not guaranteed to align with human words, but they often capture stable dimensions of variation. Word2vec and GloVe are not just engineering tricks; they sit in the same conceptual family of finding compressed latent structure in distributional data.

Noosphere's geometric methods should be read in that tradition. They assume that abstract distinctions are not purely symbolic; they leave statistical structure in a representational space.

15.3 The Difference Between Similarity and Relation

Similarity asks whether two vectors are near each other. Relation asks whether a transformation from one vector to the other has meaning. In knowledge graph embedding, TransE models a relation as something like:

$$h + r \approx t.$$

In contradiction geometry, the interesting object is often not a or b alone, but a transformed difference such as $b' - a$. In representation engineering, the interesting object may be a direction that shifts model behavior when added to or subtracted from an activation.

This distinction is the bridge between embeddings as retrieval tools and embeddings as reasoning substrates. Noosphere is trying to stand on the second side of that bridge, where transformations and directions matter.

15.4 High-Dimensional Geometry Has Traps

High-dimensional spaces behave differently from the two- or three-dimensional spaces used in visual intuition. Distances concentrate. Random vectors can be nearly orthogonal. Sparse coordinates can dominate norms. A direction that is meaningful in one model layer may be meaningless in another model or even in a different normalization regime.

This is why any geometric claim needs:

- a fixed embedding model,
- a fixed text span policy,
- a fixed normalization policy,
- a fixed metric,
- a calibration set,
- a benchmark with negative and positive controls, and
- cross-model replication before generalization.

15.5 Hoyer Sparsity

Hoyer sparsity compares the L_1 and L_2 norms of a vector. Intuitively, a vector is sparse when a few coordinates carry most of its mass. In Noosphere’s contradiction geometry, a sparse difference vector is treated as evidence that two claims differ along concentrated dimensions rather than diffusely across topic.

The hypothesis is reasonable enough to test, but it is not self-validating. A sparse difference can arise from artifacts: short text, tokenization, embedding model quirks, domain-specific vocabulary, or a threshold calibrated on the wrong distribution.

15.6 Householder Reflection

A Householder reflection maps a vector across a hyperplane. Given a unit vector d , the reflection of b is:

$$b' = b - 2(b \cdot d)d.$$

In the local contradiction engine, d is an estimated contradiction direction. The reflected vector b' is then compared with a through Hoyer sparsity. The conceptual bet is that contradiction has a direction-like structure, and that reflecting across it exposes a sharper difference signal.

The local Householder ablation is the right kind of methodological pressure: keep the step only if it survives powered comparisons. The existing ablation did not have label-level power under the frozen threshold, so its honest conclusion is neither victory nor refutation.

16 Deep Dive III: Truth, Linear Representations, and Steering

16.1 Linear Representation Hypothesis

The Linear Representation Hypothesis says, roughly, that high-level concepts can often be represented as linear directions in a model’s representation space. This does not mean every concept has one universal

axis. It means that, under some conditions, a linear probe or direction can recover a concept well enough to be scientifically and operationally meaningful.

For Noosphere, this is the external literature closest to the Quintin Hypothesis. If truth, falsehood, contradiction, uncertainty, or coherence can be directional, then a geometry-aware intellectual model is plausible. If those properties are distributed, nonlinear, layer-specific, or heavily entangled, then a naive direction detector will fail.

16.2 Truth Directions

The Geometry of Truth paper and related work ask whether factual truth has a linear structure in LLM activation space. Contrast-consistent search asks whether a direction can be found from logical constraints without labels. Inference-Time Intervention asks whether modifying truth-related directions can change model answers.

The key distinction is between **detecting** a signal and **earning** a truth claim. A direction correlated with truth on a dataset may fail under distribution shift, adversarial examples, ambiguous statements, normative claims, or claims requiring unavailable world knowledge. Noosphere's claims are often philosophical, strategic, or methodological, not only factual quiz items.

16.3 Representation Engineering

Representation engineering generalizes the workflow:

1. Define a contrast or behavior.
2. Collect examples or use unsupervised constraints.
3. Find a direction or subspace.
4. Test whether the direction predicts held-out cases.
5. Intervene and see whether behavior changes as predicted.
6. Audit side effects.

Noosphere can borrow this discipline. A "coherence direction" should not be declared because it sounds plausible. It should be operationalized, tested against counterexamples, calibrated, ablated, and versioned.

16.4 Superposition as a Warning

Superposition says a model can pack more features into a representation than it has clean dimensions for. A single neuron or direction can be polysemantic. That means a direction that looks like "truth" may also encode topic, genre, certainty, sentiment, or dataset artifacts. Sparse autoencoders attempt to recover more interpretable features, but they introduce their own training and interpretation assumptions.

For Noosphere, superposition is the strongest warning against over-literal geometry. If features are entangled, the system needs humility: multiple signals, human review, model-version tracking, and refusal to treat one scalar as metaphysical truth.

17 Deep Dive IV: Extraction Engineering

17.1 The Extraction Contract

A good extractor is not a summarizer. It is a contract-bound transformation from text to structured propositions. The contract should define:

- the allowed claim types,
- the unit of extraction,
- speaker attribution,
- endorsement versus mention,
- hedges and uncertainty,
- evidence pointers,
- source span preservation,
- refusal conditions, and
- schema validation rules.

The local `ClaimExtractor` does several of these correctly. It demands JSON, names claim types, carries confidence hedges, and tries to distinguish author assertions from external claims. The principle extractor adds decision rule shape, domain of applicability, quantifiable proxies, and decision examples.

17.2 Chunking and Span Selection

Chunking determines what context the model sees. Too-small chunks lose the premise that makes a claim intelligible. Too-large chunks invite summarization and conflation. A rigorous chunking policy should preserve:

- sentence boundaries where possible,
- paragraph-level context,
- source ids and offsets,
- speaker turns,
- quoted versus authorial text, and
- enough surrounding text for evidence pointers to be audited.

For a conversation analyzer such as *Dialectic*, this becomes harder because the source stream is temporal, noisy, and speaker-attributed. Live claims may be repaired or qualified later in the conversation.

17.3 Coreference and Entity Hygiene

Knowledge graphs degrade quickly when entity resolution is poor. If "OpenAI," "the company," "they," and "the lab" become separate nodes when they refer to the same entity, the graph fragments. If two similarly named entities are merged incorrectly, the graph fabricates support or contradiction.

Coreference resolution and entity linking therefore belong in the intellectual capital stack, not in a low-level NLP appendix. They determine whether principles and claims attach to the right objects.

17.4 Relation Extraction

A graph edge is stronger than a sentence-level label. Extracting a `supports` edge says not merely that two claims are similar, but that one is evidence or reason for another. Extracting a `contradicts` edge says that the claims cannot jointly hold under the intended interpretation. These edges should be harder to create than tags.

The best pattern is a staged extractor:

1. extract candidate claims,

2. preserve source spans,
3. classify candidate relations,
4. ask for evidence snippets,
5. validate that evidence snippets occur in the source,
6. run independent checks such as NLI or argumentation,
7. mark low confidence rather than forcing a definitive edge.

17.5 Prompting and RAG

Prompting surveys matter because many practical extraction prompts are crude versions of known strategies: decomposition, self-consistency, tool use, scratchpad reasoning, and retrieval grounding. RAG matters because an LLM's answer should be conditioned on the corpus slice, not on latent parametric memory, when the system is claiming to reason from the firm's intellectual capital.

The trace should therefore distinguish:

- what came from the retrieved corpus,
- what came from the model's general knowledge,
- what was inferred from graph structure,
- what was computed from embeddings, and
- what was generated as narrative explanation.

18 Deep Dive V: Graphs, Ontologies, and Cascade Reasoning

18.1 Property Graph vs. Formal Ontology

A property graph stores nodes, edges, and attributes. A formal ontology adds typed classes, constraints, subsumption, and satisfiability. Noosphere today is closer to a property graph with methodological discipline than to a full description-logic ontology. That is appropriate for speed, but the distinction should stay explicit.

When Noosphere says a principle "instantiates" another, or a claim "depends on" another, those words should have operational meaning. Otherwise the graph becomes decorated prose.

18.2 Edge Semantics

Important edge types mean different things:

Supports. One node raises confidence in another.

Refutes or contradicts. One node lowers confidence or creates incompatibility.

Depends on. The target requires the source as a premise or component.

Refines. The source makes the target more precise.

Specializes or generalizes. The edge moves between abstraction levels.

Extracted from. The edge records provenance from artifact to proposition.

These distinctions matter because revision should propagate differently across them. A retracted support weakens a claim. A retracted dependency may undermine it. A retracted example may only narrow scope.

18.3 Cycle Handling

Real belief graphs can contain cycles. A theory can support a method that supports a result that strengthens the theory. Cycles are not automatically bad, but some computations require acyclic structure. The local Bayesian skeleton handles this by projecting truth-valued nodes and deterministically dropping edges that would close cycles, while reporting which edges were dropped.

That is the right design instinct: do not pretend cycles do not exist, and do not let hidden cycle-breaking silently shape posterior probabilities.

18.4 Knowledge Graph Embeddings

KG embeddings translate graph structure into vector geometry. They can support link prediction, entity alignment, and reasoning over incomplete graphs. But they optimize graph reconstruction tasks, not necessarily truth or philosophical validity.

Noosphere can use KG embeddings as one signal among several. It should not let link-prediction confidence become intellectual confidence without calibration.

19 Deep Dive VI: Coherence, Probability, and Critique

19.1 Coherence as Constraint Satisfaction

Thagard's computational coherentism models coherence as satisfying positive and negative constraints across a network. This maps naturally to Noosphere: support edges are positive constraints, contradiction or refutation edges are negative constraints, and a coherent interpretation is one that satisfies as many important constraints as possible.

The appeal is obvious. It turns philosophical coherence into an optimization problem. The danger is equally obvious. Optimization requires weights, and weights smuggle in assumptions. If the graph underweights a devastating objection or overweights a favorite principle, the resulting "coherence" is only as honest as the weight assignment.

19.2 Coherence Is Not Probability

Bovens and Hartmann's warning is central: more coherence does not, in general, mean higher probability of truth. A tightly mutually supporting set of false beliefs can be coherent. A messy set of true reports from noisy sources can be less coherent. Coherence can support justification, but only under conditions that connect the belief system to reliable sources and evidence.

Theseus already gestures toward this by separating source credibility, cascade edges, Bayesian derived views, and calibration. The system should keep those separations sharp. A coherence score should answer "how well does this fit the corpus and constraints?" A probability should answer "given this model and this evidence, how likely is the proposition?" A decision trace should answer "given our rules and costs, what should we do?"

19.3 Paraconsistency and Non-Explosion

A practical knowledge graph cannot reject every contradiction instantly. Conflicts may be unresolved, source-dependent, time-dependent, or domain-specific. Paraconsistent logic provides one family of tools for preserving reasoning in the presence of contradictions without allowing arbitrary conclusions.

The engineering equivalent is to keep contradictions local. A contradiction edge should trigger review, lower confidence, veto decisions, or split contexts. It should not corrupt the whole graph.

19.4 Belief Revision

AGM-style belief revision asks how a rational belief set should change when new information arrives. Noosphere’s version is operational: a source retracts, a forecast resolves, a method fails, or an external critique lands. The system must update confidence, propagate effects, and preserve an audit trail.

The revision problem is not merely mathematical. It is institutional. A firm that cannot lower confidence in its favorite principles after contrary evidence does not have an intellectual capital model. It has a memory palace.

20 Deep Dive VII: Verification and Method Discipline

20.1 What Must Be Versioned

For Noosphere outputs to be scientifically interpretable, the following must be versioned or recorded:

- source document hash,
- chunking policy,
- extraction prompt and schema,
- LLM provider and model version,
- embedding model and dimensionality,
- normalization and distance metric,
- graph edge schema,
- coherence method versions,
- calibration bundle,
- benchmark dataset hash,
- rule graph version, and
- generated explanation prompt.

Without these, a future result cannot be reproduced or compared. With them, failure can be localized.

20.2 What Counts as Evidence for the Quintin Hypothesis

Weak evidence:

- a few compelling anecdotes,
- high cosine similarity examples,
- model-written explanations that sound plausible,
- one benchmark with no ablation,
- a scalar score without calibration.

Stronger evidence:

- frozen datasets with held-out splits,
- negative controls and random baselines,
- cosine-only and NLI-only baselines,

- cross-domain and cross-model replication,
- threshold calibration by embedder,
- powered ablations of Householder and direction-learning steps,
- confidence intervals, not only point estimates,
- failure mode catalogs tied to examples.

The local benchmark documents are valuable because they use the stronger style even when the result is uncomfortable.

20.3 Decision-Trace Verification

Before trusting an investable output, verify:

1. the market or decision input was parsed correctly;
2. the retrieved corpus slice is visible and source-bound;
3. active principles are quoted or summarized faithfully;
4. every metric has a method name and version;
5. every low-confidence flag is surfaced;
6. vetoes are evaluated before stake sizing;
7. the rule graph is stored as data;
8. the narrative summary is generated from trace fields;
9. any live action has operator authorization;
10. resolved outcomes flow back into calibration.

21 Exercises for Rebuilding Understanding

21.1 Exercise 1: Re-derive Attention

Take a five-word sentence. Assign each token a toy vector of three numbers. Choose tiny query, key, and value matrices. Compute attention scores, apply a softmax, and produce one updated token vector. The numbers will be artificial, but the exercise forces you to understand attention as routing rather than magic.

21.2 Exercise 2: Show the Cosine Paradox

Write two pairs of sentences:

- same topic, opposite truth condition;
- different topic, same sentiment or form.

Embed them with the same model. Compare cosine similarity and then inspect the difference vector. The goal is not to prove Noosphere’s method. The goal is to feel why similarity and contradiction are different questions.

21.3 Exercise 3: Build a Tiny Cascade

Create five claims:

- one central conclusion,
- two supporting claims,
- one objection,
- one qualification.

Draw the graph. Then retract one support and ask how confidence should change. If your answer is not edge-type-dependent, your graph semantics are too weak.

21.4 Exercise 4: Coherence vs. Truth

Construct two belief sets. Make one highly coherent but false, and one messy but mostly true. This exercise is the antidote to equating coherence with truth. Then ask what extra evidence would let coherence become a useful proxy for justification.

21.5 Exercise 5: Write a Decision Trace

Take a simple prediction-market question. Define the event, corpus slice, active principles, three metrics, two vetoes, and a final decision. Then write the prose summary only after the trace is complete. This mirrors the discipline Theseus needs if intellectual capital is to become accountable action.

22 Source Inventory

The following table records what was cached and how each source was used. Rows marked as HTML are saved as cleaned text in the cache; rows marked as PDF include page counts when PyMuPDF could read the file.

Domain	Source	Status	Reader use
LLM foundations	3Blue1Brown Neural Networks series	cached-html	Conceptual visual foundation for networks, gradients, attention, and transformers.
LLM foundations	Jay Alammar, The Illustrated Transformer	cached-html	First-pass architecture map: embeddings, attention, encoder/decoder blocks.
LLM foundations	Stephen Wolfram, What Is ChatGPT Doing?	cached-html	Nontechnical explanation of next-token prediction, embeddings, and generated structure.
LLM foundations	Andrej Karpathy, Let's build GPT	metadata-only	Implementation-oriented mental model for attention, blocks, loss, and training.
LLM foundations	Jurafsky and Martin, Speech and Language Processing	cached-html	Textbook spine for neural NLP, transformers, embeddings, extraction, and QA.
LLM foundations	Vaswani et al., Attention Is All You Need	cached-pdf; 15 pp	Original transformer architecture; read for attention, positional encoding, and residual blocks.
LLM foundations	Lena Voita, NLP Course	cached-html	Careful course notes for embeddings, attention, transfer learning, and analysis.

Domain	Source	Status	Reader use
Scaling and alignment	Kaplan et al., Scaling Laws for Neural Language Models	cached-pdf; 30 pp	Why model behavior changes predictably with compute, data, and parameter count.
Scaling and alignment	Hoffmann et al., Training Compute-Optimal Large Language Models	cached-pdf; 36 pp	Chinchilla correction: data/parameter ratio matters, not just parameter count.
Scaling and alignment	Ouyang et al., Training Language Models to Follow Instructions	cached-pdf; 68 pp	Instruction tuning and RLHF; explains why production LLMs are not raw predictors.
Scaling and alignment	Bai et al., Constitutional AI	cached-pdf; 34 pp	AI-feedback alignment and model priors relevant to Claude-backed extraction.
Interpretability	Elhage et al., A Mathematical Framework for Transformer Circuits	cached-html	Mechanistic decomposition of transformer computations into circuits.
Interpretability	Elhage et al., Toy Models of Superposition	cached-html	Why features can be polysemantic and why a clean coherence direction may not exist.
Interpretability	Templeton et al., Scaling Monosemanticity	cached-html	Sparse autoencoders and interpretable features in a production-scale model.
Interpretability	Olah et al., Distill Circuits thread	cached-html	Methodological ancestor of circuit-level interpretability.
Interpretability	Belinkov and Glass, Analysis Methods in Neural Language Processing	cached-pdf; 27 pp	Taxonomy of probing and behavioral analysis methods.
Embedding geometry	3Blue1Brown Essence of Linear Algebra	cached-html	Geometric intuition for vectors, bases, eigenvectors, and transformations.
Embedding geometry	MIT OCW 18.06 Linear Algebra	cached-html	Formal linear algebra backbone for embeddings and transformations.
Embedding geometry	Jay Alammar, The Illustrated Word2vec	cached-html	Visual bridge from words to distributed vectors.
Embedding geometry	Mikolov et al., Efficient Estimation of Word Representations	cached-pdf; 12 pp	Historical source for learned word embeddings as semantic objects.
Embedding geometry	Jurafsky and Martin, Chapter 6: Vector Semantics and Embeddings	cached-pdf; 27 pp	PPMI, SVD, word2vec, GloVe, and embedding evaluation.
Embedding geometry	Pennington et al., GloVe	cached-pdf; 12 pp	Co-occurrence matrix factorization view of embeddings.
Embedding geometry	Reimers and Gurevych, Sentence-BERT	cached-pdf; 11 pp	Sentence-level embeddings, directly relevant to claim and principle vectors.
Embedding geometry	Cuturi and Peyre, Computational Optimal Transport	cached-pdf; 209 pp	Distribution-level geometry for comparing clouds of embeddings.
Embedding geometry	Bronstein et al., Geometric Deep Learning	cached-pdf; 160 pp	Unifies learning under symmetry across grids, graphs, manifolds, and transformers.
Embedding geometry	Fefferman et al., Testing the Manifold Hypothesis	cached-pdf; 48 pp	Mathematical version of the claim that high-dimensional data lies on low-dimensional structure.
Representation geometry	Park, Choe, and Veitch, Linear Representation Hypothesis	cached-pdf; 24 pp	Formal backbone for treating concepts as linear directions in LLM representations.

Domain	Source	Status	Reader use
Representation geometry	Marks and Tegmark, Geometry of Truth	cached-pdf; 22 pp	Empirical truth directions in LLM activation space.
Representation geometry	Burns et al., Discovering Latent Knowledge without Supervision	cached-pdf; 28 pp	Contrast-consistent search and logical consistency directions.
Representation geometry	Zou et al., Representation Engineering	cached-pdf; 56 pp	Practical direction-finding and direction-steering toolkit.
NLP extraction	Manning et al., Computational Linguistics and Deep Learning	cached-pdf; 14 pp	Deep learning transition in NLP.
NLP extraction	Advanced NLP with spaCy	cached-html	Classical pipeline implementation: tokenization, NER, dependency parsing, matching.
NLP extraction	Natural Language Processing with Python	cached-html	Friendly introduction to rule-based and statistical NLP pipelines.
NLP extraction	Jurafsky and Martin, Chapter 18: Information Extraction	cached-pdf; 26 pp	NER, relation extraction, event extraction, temporal expressions.
NLP extraction	Devlin et al., BERT	cached-pdf; 16 pp	Pretrained bidirectional encoder behind modern extraction systems.
NLP extraction	Lee et al., End-to-end Neural Coreference Resolution	cached-pdf; 10 pp	Coreference as a prerequisite for clean graph nodes.
NLP extraction	Cabot and Navigli, REBEL	cached-pdf; 12 pp	Relation extraction by sequence generation.
NLP extraction	Wang et al., Neural Relation Extraction Survey	cached-pdf; 13 pp	Survey of relation extraction approaches before instruction-tuned LLMs.
NLP extraction	Wadhwa et al., Information Extraction from LLMs	cached-pdf; 22 pp	Zero/few-shot LLM extraction behavior and failure modes.
NLP extraction	Qiao et al., Reasoning with Language Model Prompting	cached-pdf; 24 pp	Prompting survey: decomposition, chain-of-thought, self-consistency, tools.
NLP extraction	Lewis et al., Retrieval-Augmented Generation	cached-pdf; 19 pp	Retrieval-grounded generation and knowledge-intensive tasks.
NLP extraction	Stab and Gurevych, Parsing Argumentation Structures	cached-pdf; 42 pp	Claims, premises, support relations, and argument mining.
Knowledge graphs	Hogan et al., Knowledge Graphs	cached-pdf; 135 pp	Knowledge graph handbook: data models, identity, quality, refinement.
Knowledge graphs	NetworkX tutorial	cached-html	Practical graph representation and traversal operations used in Noosphere.
Knowledge graphs	Diestel, Graph Theory	cached-html	Formal graph-theory reference. Full downloadable PDF was not assumed from mirrors.
Logic and coherence	SEP, Coherentist Theories of Epistemic Justification	cached-html	Philosophical target: coherence as justification, not automatically truth.
Logic and coherence	SEP, Coherence Theory of Truth	cached-html	Distinguishes coherence-as-truth from coherence-as-justification.
Knowledge graphs	Bordes et al., TransE	cached-pdf; 9 pp	Knowledge-graph embeddings as translation in vector space.
Knowledge graphs	Wang et al., Knowledge Graph Embedding Survey	cached-html	KG embedding vocabulary and model taxonomy; page cached, full IEEE PDF not public here.
Logic and coherence	SEP, Paraconsistent Logic	cached-html	Reasoning without explosion under contradiction.

Domain	Source	Status	Reader use
Logic and coherence	SEP, Mereology	cached-html	Part-whole relations for principle distillation and claim composition.
Logic and coherence	SEP, Logic of Belief Revision	cached-html	Formal belief update and revision.
Knowledge graphs	Nickel et al., Relational Machine Learning for Knowledge Graphs	cached-pdf; 23 pp	Relational learning, latent-feature models, and graph-feature models.
Representation geometry	Lin et al., TruthfulQA	cached-pdf; 39 pp	Truthfulness benchmark for models that mimic human falsehoods.
Representation geometry	Li et al., Inference-Time Intervention	cached-pdf; 80 pp	Intervening on attention-head directions correlated with truthfulness.
Logic and coherence	Thagard, Coherence as Constraint Satisfaction	cached-pdf; 24 pp	Computational coherentism as network constraint satisfaction; author-hosted PDF cached.
Interpretability	Olsson et al., In-context Learning and Induction Heads	cached-html	Circuit-level explanation of a concrete in-context learning behavior.
Math primers	Blitzstein and Hwang, Introduction to Probability / Stat 110	cached-html	Probability prerequisite for calibration, Bayesian updates, and uncertainty.
Math primers	MIT OCW 18.02 Multivariable Calculus	cached-html	Calculus prerequisite for gradients and optimization.
Math primers	MacKay, Information Theory, Inference, and Learning Algorithms	cached-pdf; 640 pp	Information theory and probabilistic inference primer.
Math primers	Boyd and Vandenberghe, Convex Optimization	cached-pdf; 714 pp	Optimization primer for training objectives and convex geometry.
KG and LLM frontier	Pan et al., Unifying Large Language Models and Knowledge Graphs	cached-pdf; 28 pp	Current roadmap for KG-enhanced LLMs, LLM-augmented KGs, and synergized systems.
KG and LLM frontier	Bian, LLM-empowered Knowledge Graph Construction	cached-pdf; 14 pp	Recent survey of LLM-based ontology engineering, extraction, and fusion.
KG and LLM frontier	Pan et al., Large Language Models and Knowledge Graphs	cached-pdf; 30 pp	Position paper on explicit KGs and parametric knowledge in LLMs.
Theseus local	Theseus README	local	Defines Theseus as principle extraction, algorithm execution, and accountable bets.
Theseus local	Algorithmized Decision Making	local	Defines traces, rule graphs, investable outputs, and metric contracts.
Theseus local	Claim extractor implementation	local	Shows the JSON claim/principle extraction contract and external-claim filtering.
Theseus local	Embedding pipeline implementation	local	Shows how source text is embedded and written into the canonical store.
Theseus local	Canonical contradiction engine	local	Householder reflection, Hoyer sparsity, reliability calibration, and limits.
Theseus local	Bayesian Belief Layer	local	Derived Bayesian network over the cascade graph.
Theseus local	QH Benchmark v1 Results	local	Empirical warning: firm contradiction geometry underperformed on accuracy in the first run.

Domain	Source	Status	Reader use
Theseus local	Householder Ablation	local	Empirical warning: the ablation was zero-power under the frozen threshold.

23 Closing Synthesis

The deepest question behind Noosphere is not whether LLMs can write plausible summaries. They can. The question is whether a system can turn interpreted text into a durable, auditable, and empirically disciplined model of intellectual capital.

The strongest version of the architecture is a hybrid:

- LLMs do high-recall linguistic work, but schema validation and provenance discipline prevent them from silently authoring the knowledge base.
- Embeddings supply soft geometry, but benchmarked calibration prevents cosine or sparsity from being mistaken for logic.
- Graphs make support, contradiction, and dependency inspectable, but probabilistic and paraconsistent layers prevent the graph from pretending to be cleaner than it is.
- Decision traces turn principles into accountable computations, while market or forecast outcomes provide calibration pressure.

The source literature makes this project plausible. The local benchmark documents make it intellectually honest: the geometric hypothesis is promising, but not yet vindicated. The rigorous path forward is not to weaken the claim into a vague metaphor. It is to keep the claim sharp enough that benchmarks, ablations, source retractions, and failed forecasts can actually wound it.